

2.2 · Discussion — The Environmental Cost of Creating LLMs

Week 2 · Tuesday Jul 21 — Data, Environment, and Human-Centered Sustainable AI

Generative AI & Sustainability · Munich Summer School 2026



CAL POLY



Today's paper

Holistically Evaluating the Environmental Impact of Creating Language Models

Morrison, Na, Fernandez, Dettmers, Strubell, Dodge — ICLR 2025 (spotlight) ·
arXiv:2503.05804

The most complete public accounting of what it costs — in carbon **and** water — to create a family of language models, including the parts developers almost never report.

Feel free to reference your **admission ticket**.

How they measured it

They accounted for **four cost buckets**, not just the one everyone reports:

hardware manufacturing (embodied) · development · final training · deployment/inference

- Measured **real GPU power at sub-second intervals** on the actual training cluster — prior work just assumes 100% of the chip's rated draw
- GPU-only, so every number is a **lower bound**
- Subject: the open **OLMo** model family, 20M → 13B parameters, two US data centers

What it cost

Take **one model** — **OLMo 2 7B** — the final training run alone:

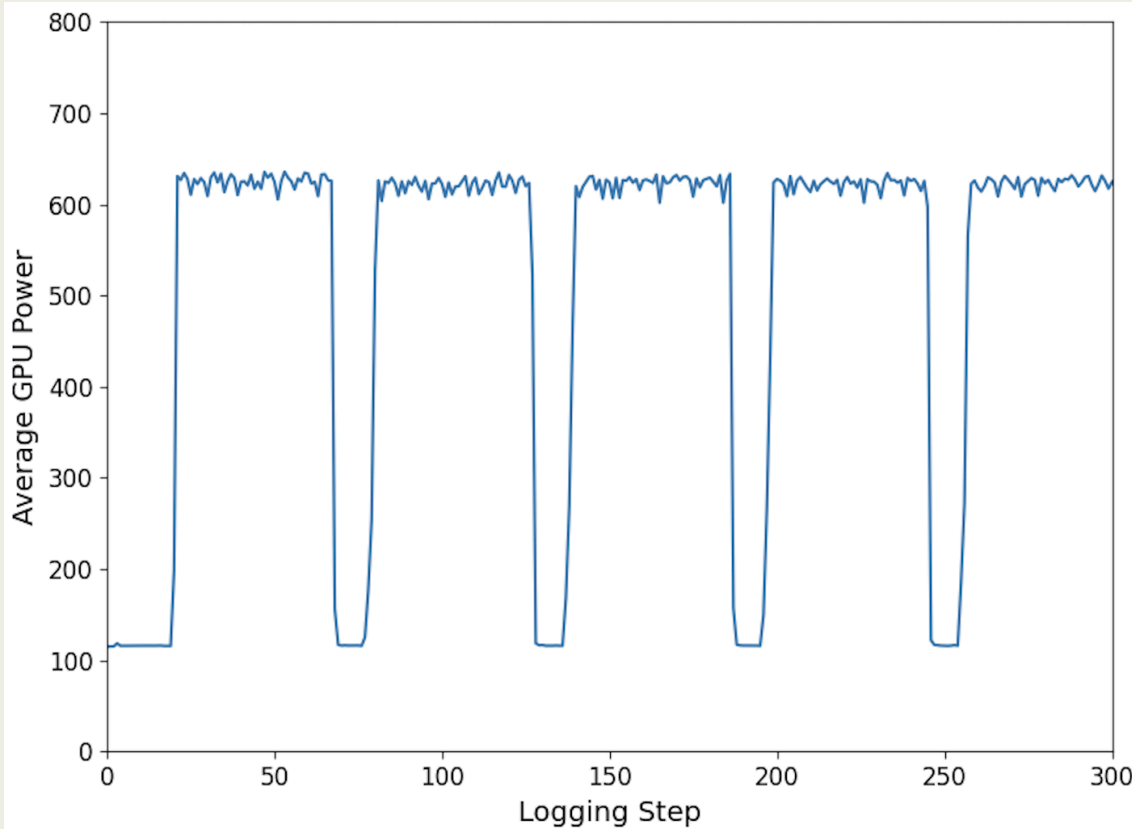
Power	Carbon	Water
157 MWh	52 tCO ₂ eq	202 kL
—	≈ 10 yrs of one US home's electricity	≈ 1 yr 9 mo of one person's water

- This is only the **final run** — development (tuning, discarded runs) sits on top and is almost never reported
- **Same model, clean grid**: an earlier OLMo 7B trained on hydroelectric power emitted ≈ **0** — the number is about the *electricity*, not the model
- All the water is **off-site** (closed-loop cooling → 0 L on-site); the electricity itself consumes it
- Across the whole OLMo family this compounds to **493 tCO₂eq** and **2,769 kL** of water

What it means

- Standard disclosures **undercount** — they hide development, embodied hardware, and water entirely
- **Inference eventually dominates**: break-even with training arrives in weeks to months at real serving scale, and Jevons' Paradox means efficient small models *increase* total demand
- The authors' call: **mandatory reporting across the whole supply chain** (EU AI Act Art. 95)

Notable image: GPU power swings



- **Pattern:** the first spike starts training; each drop saves a checkpoint
- **Magnitude:** over 600W while training (>85% of H100 maximum); just over 100W during checkpoints (~15%)
- **Grid impact:** rapid swings strain grid control at scale; parallel checkpointing and demand response reduce the strain

Why we discuss instead of lecture

On **Thursday** you will present your system & what it costs the planet.

Today you practice answering that question with evidence — arguing from a paper that measured it end to end, and defending your evaluation and reporting choices out loud.

Warm-up — think, pair, share

Which single number in this paper most changed how you think about the cost of building an LLM — and what would you have guessed before reading?

- **Think + write** on your own
- **Pair** with your neighbour and compare
- **Share** with the room

Team activity

I will split you in to groups.

The question to analyze

Which one recommendation from the paper's conclusion would matter most if it became law — for example, written into the EU AI Act?

Role	n	Job
Proponent	1–2	Pick one recommendation; make the strongest case for its impact, with paper evidence
Critic	1–2	Where it would be weak, gamed, or hard to enforce as law
Skeptic	1–2	Feasibility & cost — who pays, and would it hold beyond OLMo-scale?
Summarizer	1	Delivers the team's pick and one-line justification

- Any member may be asked follow-ups — everyone must know the whole analysis.

Report-out

Each team names its recommendation and defends why it matters most.

Bring:

- Your **strongest single piece of evidence**, with the number attached
- The **weakest point** in your own case — say it before the other team finds it
- One thing your recommendation still **would not fix**

Whole-group synthesis — back to your startups

1. Your **week-1 carbon footprint** for your NextGen pipeline: after this paper, what line items were missing?
2. Would you **voluntarily publish** numbers like these, knowing competitors won't? What would make disclosure rational?
3. You are pitching HM as a customer on Thursday. Write the **one sentence** of environmental reporting you would promise them.

Close

- One-sentence takeaway from each team
- **Hand in your admission tickets**
- Next (M3): what is *in* the data — leakage, privacy, and private ML
- **Tonight's reading:** Kim et al. — what *agents* cost to run, the other end of today's lifecycle. For tomorrow's fishbowl.